THE PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

FORM FOUR MOCK EXAMINATION ARUSHA REGIONAL MAY 2024

F4-BIOLOGY 2A PRACTICALS

MARKING SCHEME

1. i. Cold sensation 05 marks

ii. When the body temperature falls below the normal set point, cold thermoreceptors detect the change in temperature and send sensory information to the hypothalamus. The hypothalamus in return identifies the sensory impulse as cold sensation and initiate temperature generation mechanisms in the body such as shivering and increased liver metabolism. 15 marks

iii. Vasoconstricted 05 marks

iv Fever ~~05~~ 05 marks

2. (a) U – Quill feather

V – Crab

W – Cactus 01/2 @ = 2.5

X – Earthworm

Y- Tilapia fish

(b) In dry environment (desert and semi desert)

Adaption of specimen W to its environment 01/2 @ = 2.5

not cancelled!!

(i). Has extensive root system for water absorption

(ii). Has thorns which discourage herbivores from feeding on its their succulent parts

(iii). Has few numbers of stomata to minimize transpiration

(iv). Has thick and water proof cuticle to reduce evaporation

2. (c) Classification of specimens U, V, X and Y to the class level

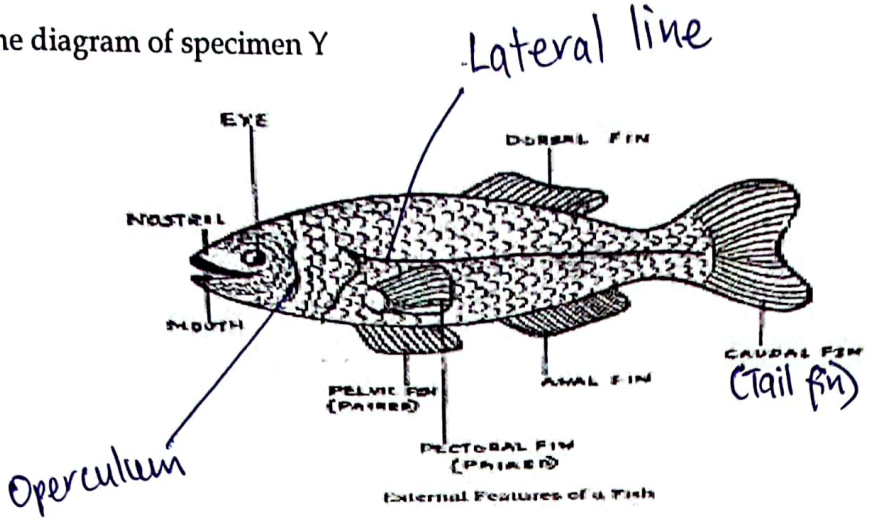
Specimens	*Kingdom*	*Phylum*	*Class*
U – (Quill feather)	Animalia	Chordata	Aves
V – (Crab)	Animalia	Arthropoda	Crustacea
X – (Earth worm)	Animalia	Annelida	Oligochaeta

01/2 @ = 02

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Y – (Tilapia fish)	Animalia	Chordata	Osteichthyes

2. (d) The diagram of specimen Y



03 Marks

2. (e). Observed structures are hairs and pores . they are both involved in the body temperature regulation.

02 marks

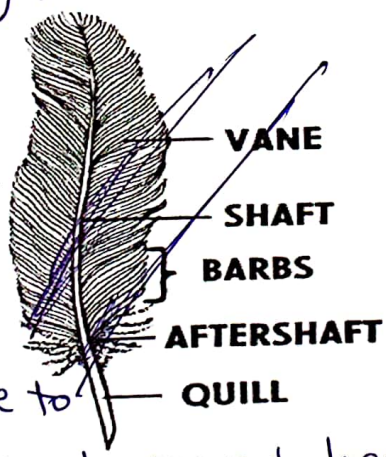
2. (f) Economic importance of specimens X are: -

(i). It increases aeration in the soil
(ii). It increases water holding capacity in the soil
(iii). It increases soil fertility as it decomposes after death

03 marks

2. (g) ~~Diagram of specimen Y~~ 02 marks

2(g) Roles of specimen 'V'.



(i) It reduces air resistance during flying
(ii) It increases the surface area for flight
(iii) It makes birds less dense (reduces weight)
(iv) It protects against mechanical damage to the skin.
(v) It enables body insulation to prevent heat loss.

2. (h) Adaptation of specimen Y

-Has fins for swimming
- Has scales for protection
- Has gills for gaseous exchange

02 Marks

-Have air bladder for bouyancy when swimming.
- Have pointed head (streamlined body) to reduce friction when ~~S~~swimming
- Has lateral line which help the fish to sense vibrations and water pressure.

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